Practical - 13

Practical Title : PHP JSON API and AJAX/Fetch-Based CRUD without Page Reload

Problem Definition: Create simple PHP JSON API endpoints for events or students and consume them using Fetch API. Perform list, add, update, and delete operations asynchronously without full page reload.

Theory

1. **Do PHP endpoints return valid JSON responses?**
   Answer: PHP endpoints return valid JSON responses for successful and failed requests.

2. **Are GET and POST requests handled properly?**
   Answer: GET and POST requests are handled properly to retrieve and submit data.

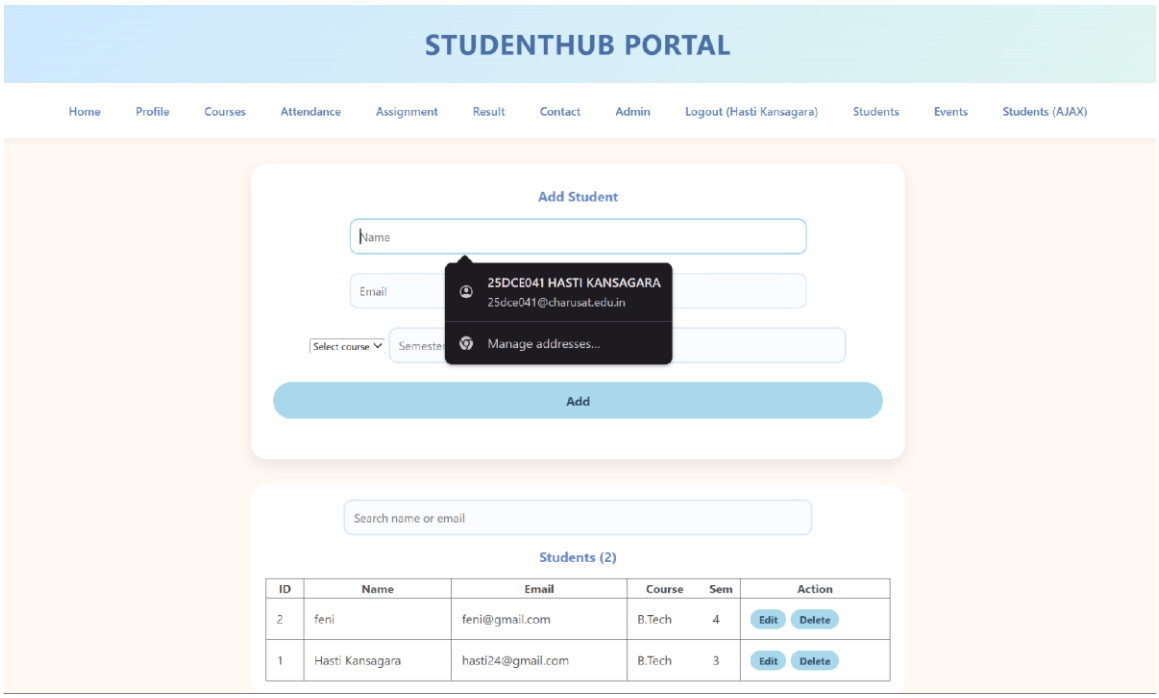
3. **Are loading, success, and error states shown on the frontend?**
   Answer: Loading, success, and error messages are displayed on the frontend.

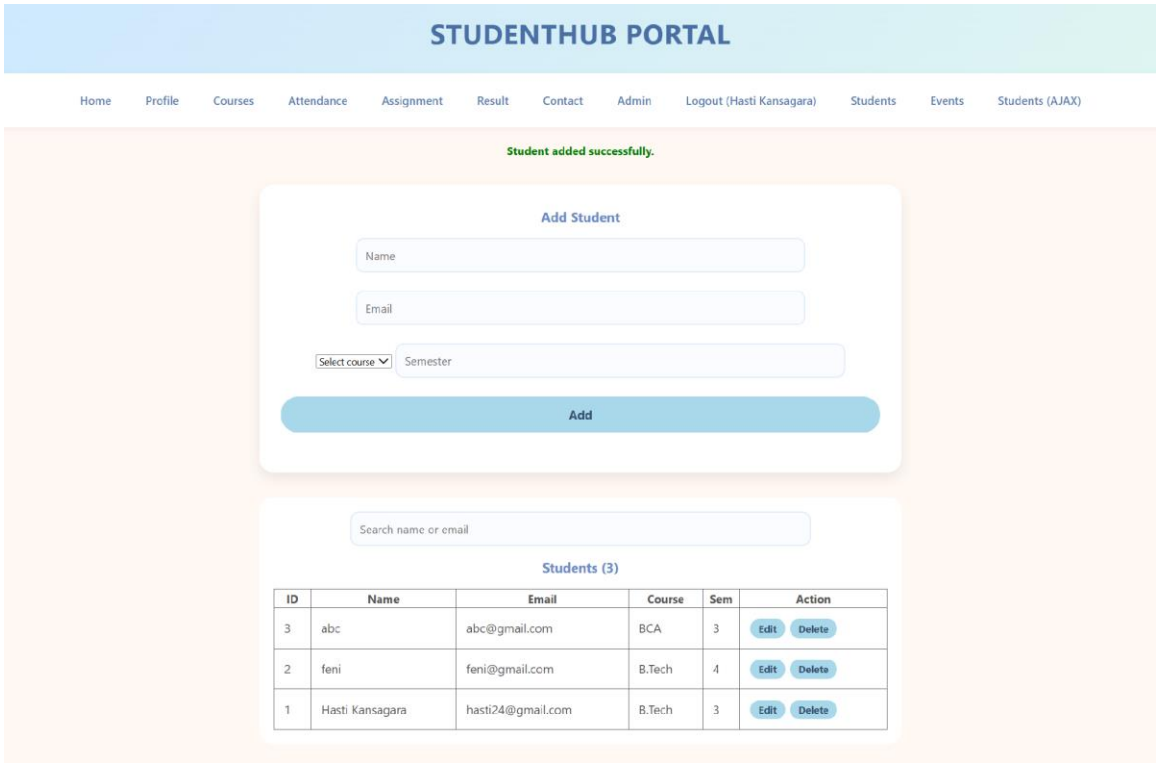
4. **Is input validation performed before database operations?**
   Answer: Input validation is performed before executing database operations to ensure data accuracy and security.

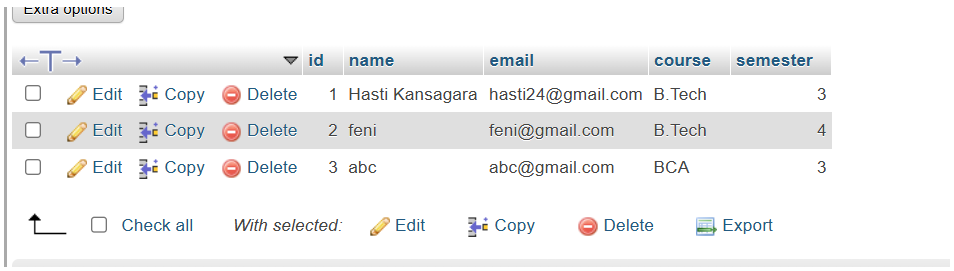
Implementation :

- Created PHP endpoints to return valid JSON responses.
- Handled GET requests to retrieve data and POST requests to submit data.
- Added loading indicators and success and error messages on the frontend.
- Implemented input validation before performing database operations.
- Connected the frontend with PHP endpoints to exchange data dynamically.

A. Evidence of Work (Screenshots)







B. GitHub Repository Link:

https://github.com/25dce049-ui/student-hub

C. Challenges Faced:

- Generating valid JSON responses from PHP endpoints.
- Handling GET and POST requests correctly.
- Displaying loading, success, and error states on the frontend.
- Validating user input before database operations.
- Managing communication between the frontend, PHP, and MySQL database.

D. Learning Outcomes:

- Learned to create PHP endpoints that return JSON responses.
- Understood how to handle GET and POST requests.
- Learned to display loading indicators and success/error messages.
- Gained knowledge of input validation and secure database operations.
- Understood how to exchange data dynamically between the frontend and backend.

Faculty Remarks: **(Keep this blank for faculty)**